Model for "O1"

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General Condition

- Basis type: `lgs`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: $[A_1, A_2, E, T_1, T_2]$
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: $[A_1, A_2, E, T_1, T_2]$
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: $[A_1, A_2, E, T_1, T_2]$
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: $[A_1, A_2, E, T_1, T_2]$
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

Group and Unit Cell

- Group: SG No. 207 O^1 $P432$ [cubic]
- Associated point group: PG No. 207 O 432 [cubic]
- Unit cell:
  $a =$ 1.00000, $b =$ 1.00000, $c =$ 1.00000, $\alpha = 90.0$, $\beta = 90.0$, $\gamma = 90.0$
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ [1.00000, 0.00000, 0.00000]
  $\boldsymbol{a}_2 =$ [0.00000, 1.00000, 0.00000]
  $\boldsymbol{a}_3 =$ [0.00000, 0.00000, 1.00000]

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1\|0\}$	2	$\{2_{001}\|0\}$	3	$\{2_{010}\|0\}$	4	$\{2_{100}\|0\}$	5	$\{3^{+}_{111}\|0\}$
6	$\{3^{+}_{-11-1}\|0\}$	7	$\{3^{+}_{1-1-1}\|0\}$	8	$\{3^{+}_{-1-11}\|0\}$	9	$\{3^{-}_{111}\|0\}$	10	$\{3^{-}_{1-1-1}\|0\}$
11	$\{3^{-}_{-1-11}\|0\}$	12	$\{3^{-}_{-11-1}\|0\}$	13	$\{2_{110}\|0\}$	14	$\{2_{1-10}\|0\}$	15	$\{4^{-}_{001}\|0\}$
16	$\{4^{+}_{001}\|0\}$	17	$\{4^{-}_{100}\|0\}$	18	$\{2_{011}\|0\}$	19	$\{2_{01-1}\|0\}$	20	$\{4^{+}_{100}\|0\}$
21	$\{4^{+}_{010}\|0\}$	22	$\{2_{101}\|0\}$	23	$\{4^{-}_{010}\|0\}$	24	$\{2_{-101}\|0\}$		

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{G}_0(A_1)$	A_1	0	G, M	-	-	1
2	$\mathbb{Q}_0(A_1)$	A_1	0	Q, T	-	-	1
3	$\mathbb{G}_4(A_1)$	A_1	4	G, M	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$
4	$\mathbb{Q}_4(A_1)$	A_1	4	Q, T	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$
5	$\mathbb{G}_3(A_2)$	A_2	3	G, M	-	-	$\sqrt{15}xyz$
6	$\mathbb{Q}_3(A_2)$	A_2	3	Q, T	-	-	$\sqrt{15}xyz$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
7	$\mathbb{G}_{2,1}(E)$	E	2	G, M	-	1	$\frac{-x^2-y^2+2z^2}{2}$
8	$\mathbb{G}_{2,2}(E)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
9	$\mathbb{Q}_{2,1}(E)$	E	2	Q, T	-	1	$\frac{-x^2-y^2+2z^2}{2}$
10	$\mathbb{Q}_{2,2}(E)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
11	$\mathbb{G}_{4,1}(E)$	E	4	G, M	-	1	$\frac{\sqrt{15}(-x^4+12x^2y^2-6x^2z^2-y^4-6y^2z^2+2z^4)}{12}$
12	$\mathbb{G}_{4,2}(E)$					2	$\frac{\sqrt{5}(x^4-6x^2z^2-y^4+6y^2z^2)}{4}$
13	$\mathbb{Q}_{4,1}(E)$	E	4	Q, T	-	1	$\frac{\sqrt{15}(-x^4+12x^2y^2-6x^2z^2-y^4-6y^2z^2+2z^4)}{12}$
14	$\mathbb{Q}_{4,2}(E)$					2	$\frac{\sqrt{5}(x^4-6x^2z^2-y^4+6y^2z^2)}{4}$
15	$\mathbb{Q}_{5,1}(E)$	E	5	Q, T	-	1	$\frac{3\sqrt{35}xyz(x^2-y^2)}{2}$
16	$\mathbb{Q}_{5,2}(E)$					2	$\frac{\sqrt{105}xyz(x^2+y^2-2z^2)}{2}$
17	$\mathbb{G}_{1,1}(T_1)$	T_1	1	G, M	-	1	x
18	$\mathbb{G}_{1,2}(T_1)$					2	y
19	$\mathbb{G}_{1,3}(T_1)$					3	z
20	$\mathbb{Q}_{1,1}(T_1)$	T_1	1	Q, T	-	1	x
21	$\mathbb{Q}_{1,2}(T_1)$					2	y
22	$\mathbb{Q}_{1,3}(T_1)$					3	z
23	$\mathbb{G}_{3,1}(T_1)$	T_1	3	G, M	-	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
24	$\mathbb{G}_{3,2}(T_1)$					2	$\frac{y(-3x^2+2y^2-3z^2)}{2}$
25	$\mathbb{G}_{3,3}(T_1)$					3	$\frac{z(-3x^2-3y^2+2z^2)}{2}$
26	$\mathbb{Q}_{3,1}(T_1)$	T_1	3	Q, T	-	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
27	$\mathbb{Q}_{3,2}(T_1)$					2	$\frac{y(-3x^2+2y^2-3z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
28	$\mathbb{Q}_{3,3}(T_1)$					3	$\frac{z\left(-3x^2-3y^2+2z^2\right)}{2}$
29	$\mathbb{G}_{4,1}(T_1)$	T_1	4	G, M	-	1	$\frac{\sqrt{35}yz\left(y^2-z^2\right)}{2}$
30	$\mathbb{G}_{4,2}(T_1)$					2	$\frac{\sqrt{35}xz\left(-x^2+z^2\right)}{2}$
31	$\mathbb{G}_{4,3}(T_1)$					3	$\frac{\sqrt{35}xy\left(x^2-y^2\right)}{2}$
32	$\mathbb{Q}_{4,1}(T_1)$	T_1	4	Q, T	-	1	$\frac{\sqrt{35}yz\left(y^2-z^2\right)}{2}$
33	$\mathbb{Q}_{4,2}(T_1)$					2	$\frac{\sqrt{35}xz\left(-x^2+z^2\right)}{2}$
34	$\mathbb{Q}_{4,3}(T_1)$					3	$\frac{\sqrt{35}xy\left(x^2-y^2\right)}{2}$
35	$\mathbb{Q}_{5,1}(T_1,2)$	T_1	5	Q, T	2	1	$\frac{3\sqrt{35}x\left(y^4-6y^2z^2+z^4\right)}{8}$
36	$\mathbb{Q}_{5,2}(T_1,2)$					2	$\frac{3\sqrt{35}y\left(x^4-6x^2z^2+z^4\right)}{8}$
37	$\mathbb{Q}_{5,3}(T_1,2)$					3	$\frac{3\sqrt{35}z\left(x^4-6x^2y^2+y^4\right)}{8}$
38	$\mathbb{G}_{2,1}(T_2)$	T_2	2	G, M	-	1	$\sqrt{3}yz$
39	$\mathbb{G}_{2,2}(T_2)$					2	$\sqrt{3}xz$
40	$\mathbb{G}_{2,3}(T_2)$					3	$\sqrt{3}xy$
41	$\mathbb{Q}_{2,1}(T_2)$	T_2	2	Q, T	-	1	$\sqrt{3}yz$
42	$\mathbb{Q}_{2,2}(T_2)$					2	$\sqrt{3}xz$
43	$\mathbb{Q}_{2,3}(T_2)$					3	$\sqrt{3}xy$
44	$\mathbb{G}_{3,1}(T_2)$	T_2	3	G, M	-	1	$\frac{\sqrt{15}x\left(y^2-z^2\right)}{2}$
45	$\mathbb{G}_{3,2}(T_2)$					2	$\frac{\sqrt{15}y\left(-x^2+z^2\right)}{2}$
46	$\mathbb{G}_{3,3}(T_2)$					3	$\frac{\sqrt{15}z\left(x^2-y^2\right)}{2}$
47	$\mathbb{Q}_{3,1}(T_2)$	T_2	3	Q, T	-	1	$\frac{\sqrt{15}x\left(y^2-z^2\right)}{2}$
48	$\mathbb{Q}_{3,2}(T_2)$					2	$\frac{\sqrt{15}y\left(-x^2+z^2\right)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
49	$\mathbb{Q}_{3,3}(T_2)$					3	$\frac{\sqrt{15}z\left(x^2-y^2\right)}{2}$
50	$\mathbb{G}_{4,1}(T_2)$	T_2	4	G, M	-	1	$\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$
51	$\mathbb{G}_{4,2}(T_2)$					2	$\frac{\sqrt{5}xz\left(-x^2+6y^2-z^2\right)}{2}$
52	$\mathbb{G}_{4,3}(T_2)$					3	$\frac{\sqrt{5}xy\left(-x^2-y^2+6z^2\right)}{2}$
53	$\mathbb{Q}_{4,1}(T_2)$	T_2	4	Q, T	-	1	$\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$
54	$\mathbb{Q}_{4,2}(T_2)$					2	$\frac{\sqrt{5}xz\left(-x^2+6y^2-z^2\right)}{2}$
55	$\mathbb{Q}_{4,3}(T_2)$					3	$\frac{\sqrt{5}xy\left(-x^2-y^2+6z^2\right)}{2}$
56	$\mathbb{Q}_{5,1}(T_2)$	T_2	5	Q, T	-	1	$\frac{\sqrt{105}x\left(2x^2y^2-2x^2z^2-y^4+z^4\right)}{4}$
57	$\mathbb{Q}_{5,2}(T_2)$					2	$\frac{\sqrt{105}y\left(x^4-2x^2y^2+2y^2z^2-z^4\right)}{4}$
58	$\mathbb{Q}_{5,3}(T_2)$					3	$\frac{\sqrt{105}z\left(-x^4+2x^2z^2+y^4-2y^2z^2\right)}{4}$

Basis in full matrix

Table 3: dimension = 8

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$\lvert s,\uparrow\rangle$ @A(1)	1	$\lvert s,\downarrow\rangle$ @A(1)	2	$\lvert p_x,\uparrow\rangle$ @A(1)	3	$\lvert p_x,\downarrow\rangle$ @A(1)	4	$\lvert p_y,\uparrow\rangle$ @A(1)
5	$\lvert p_y,\downarrow\rangle$ @A(1)	6	$\lvert p_z,\uparrow\rangle$ @A(1)	7	$\lvert p_z,\downarrow\rangle$ @A(1)				

Table 4: Atomic basis (orbital part only)

orbital	definition
$\lvert s\rangle$	1
$\lvert p_x\rangle$	x
$\lvert p_y\rangle$	y
$\lvert p_z\rangle$	z

SAMB: 604 (all 604)

- `A` : '`A`' site-cluster
  * bra: $\langle s,\uparrow\rvert$, $\langle s,\downarrow\rvert$
  * ket: $\lvert s,\uparrow\rangle$, $\lvert s,\downarrow\rangle$
  * wyckoff: `1a`

$\boxed{\text{z1}}$ $\mathbb{Q}_{0}^{(c)}(A_1) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(s)}(A_1)$

- `A` : '`A`' site-cluster
  * bra: $\langle s,\uparrow\rvert$, $\langle s,\downarrow\rvert$
  * ket: $\lvert p_x,\uparrow\rangle$, $\lvert p_x,\downarrow\rangle$, $\lvert p_y,\uparrow\rangle$, $\lvert p_y,\downarrow\rangle$, $\lvert p_z,\uparrow\rangle$, $\lvert p_z,\downarrow\rangle$
  * wyckoff: `1a`

$\boxed{\text{z2}}$ $\mathbb{G}_{0}^{(1,1;c)}(A_1) = \mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{0}^{(s)}(A_1)$

$\boxed{\text{z61}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$

$\boxed{\text{z62}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$

$\boxed{\text{z179}}$ $\mathbb{Q}_{1,1}^{(c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z180}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_0^{(s)}(A_1)}{3}$$

$\boxed{\text{z181}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_0^{(s)}(A_1)}{3}$$

$\boxed{\text{z182}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_0^{(s)}(A_1)}{3}$$

$\boxed{\text{z183}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_0^{(s)}(A_1)}{3}$$

$\boxed{\text{z184}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_0^{(s)}(A_1)}{3}$$

$\boxed{\text{z386}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$$

$\boxed{\text{z387}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$$

$\boxed{\text{z388}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(s)}(A_1)}{3}$$

- `A` : '`A`' site-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `1a`

$\boxed{\text{z3}}$ $$\mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$$

$\boxed{\text{z4}}$ $$\mathbb{Q}_0^{(1,1;c)}(A_1) = \mathbb{Q}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$$

$\boxed{\text{z63}}$ $$\mathbb{Q}_{2,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$\boxed{\text{z64}}$ $$\mathbb{Q}_{2,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$\boxed{\text{z65}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$

$\boxed{\text{z66}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$

$\boxed{\text{z185}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z186}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z187}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_1) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z389}}$ $\mathbb{Q}_{2,1}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z390}}$ $\mathbb{Q}_{2,2}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z391}}$ $\mathbb{Q}_{2,3}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z392}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z393}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

$\boxed{\text{z394}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$

- `A;A_001_1` : '`A`'-'`A`' bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `3a@3d`

$\boxed{\text{z5}}$ $\mathbb{Q}_{0}^{(c)}(A_1) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$$\boxed{\text{z6}}\quad \mathbb{G}_{0}^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$$\boxed{\text{z67}}\quad \mathbb{Q}_{2,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z68}}\quad \mathbb{Q}_{2,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

$$\boxed{\text{z69}}\quad \mathbb{G}_{2,1}^{(1,-1;c)}(E) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z70}}\quad \mathbb{G}_{2,2}^{(1,-1;c)}(E) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$$

$$\boxed{\text{z188}}\quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z189}}\quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z190}}\quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z395}}\quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z396}}\quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z397}}\quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

- `A;A_001_1` : ’`A`’-’`A`’ bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3a@3d`

$\boxed{\text{z7}}$ $\mathbb{Q}_0^{(c)}(A_1) = \dfrac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z8}}$ $\mathbb{Q}_0^{(1,0;c)}(A_1) = \dfrac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z9}}$ $\mathbb{G}_0^{(1,-1;c)}(A_1) = \dfrac{\sqrt{2}\left(\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{2}$

$\boxed{\text{z10}}$ $\mathbb{G}_0^{(1,1;c)}(A_1) = \mathbb{G}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z39}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_2) = \dfrac{\sqrt{2}\left(\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{2}$

$\boxed{\text{z40}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2) = \dfrac{\sqrt{3}\left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z71}}$ $\mathbb{Q}_{2,1}^{(c)}(E) = \dfrac{\sqrt{3}\left(-\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z72}}$ $\mathbb{Q}_{2,2}^{(c)}(E) = \dfrac{\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$

$\boxed{\text{z73}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E) = \dfrac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$

$\boxed{\text{z74}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E) = \dfrac{\sqrt{3}\left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z75}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E) = \dfrac{\sqrt{3}\left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z76}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E) = \dfrac{\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$

$\boxed{\text{z77}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E,a) = \dfrac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_0^{(b)}(A_1)}{2}$

$\boxed{\text{z78}}$ $$\mathbb{G}^{(1,-1;c)}_{2,2}(E,a)=\frac{\sqrt{2}\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{0}(A_1)}{2}$$

$\boxed{\text{z79}}$ $$\mathbb{G}^{(1,-1;c)}_{2,1}(E,b)=\frac{\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,1}(E)-\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,2}(E)}{2}$$

$\boxed{\text{z80}}$ $$\mathbb{G}^{(1,-1;c)}_{2,2}(E,b)=-\frac{\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,2}(E)+\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,1}(E)}{2}$$

$\boxed{\text{z81}}$ $$\mathbb{G}^{(1,1;c)}_{2,1}(E)=\frac{\sqrt{2}\mathbb{G}^{(1,1;a)}_{0}(A_1)\mathbb{Q}^{(b)}_{2,1}(E)}{2}$$

$\boxed{\text{z82}}$ $$\mathbb{G}^{(1,1;c)}_{2,2}(E)=\frac{\sqrt{2}\mathbb{G}^{(1,1;a)}_{0}(A_1)\mathbb{Q}^{(b)}_{2,2}(E)}{2}$$

$\boxed{\text{z191}}$ $$\mathbb{Q}^{(c)}_{1,1}(T_1,a)=\frac{\sqrt{3}\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{0}(A_1)}{3}$$

$\boxed{\text{z192}}$ $$\mathbb{Q}^{(c)}_{1,2}(T_1,a)=\frac{\sqrt{3}\mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{0}(A_1)}{3}$$

$\boxed{\text{z193}}$ $$\mathbb{Q}^{(c)}_{1,3}(T_1,a)=\frac{\sqrt{3}\mathbb{Q}^{(a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{0}(A_1)}{3}$$

$\boxed{\text{z194}}$ $$\mathbb{Q}^{(c)}_{1,1}(T_1,b)=\frac{-\sqrt{3}\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,1}(E)+3\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,2}(E)}{6}$$

$\boxed{\text{z195}}$ $$\mathbb{Q}^{(c)}_{1,2}(T_1,b)=-\frac{\sqrt{3}\mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,1}(E)+3\mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,2}(E)}{6}$$

$\boxed{\text{z196}}$ $$\mathbb{Q}^{(c)}_{1,3}(T_1,b)=\frac{\sqrt{3}\mathbb{Q}^{(a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{2,1}(E)}{3}$$

$\boxed{\text{z197}}$ $$\mathbb{Q}^{(1,-1;c)}_{1,1}(T_1)=\frac{3\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,1}(E)+\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)}{6}$$

$\boxed{\text{z198}}$ $$\mathbb{Q}^{(1,-1;c)}_{1,2}(T_1)=\frac{-3\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(E)+\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)}{6}$$

$\boxed{\text{z199}}$ $$\mathbb{Q}^{(1,-1;c)}_{1,3}(T_1)=-\frac{\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)}{3}$$

$\boxed{\text{z200}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z201}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z202}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z203}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_1,b) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$$

$\boxed{\text{z204}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_1,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$$

$\boxed{\text{z205}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_1,b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)}{3}$$

$\boxed{\text{z206}}$ $$\mathbb{G}_{1,1}^{(c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z207}}$ $$\mathbb{G}_{1,2}^{(c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z208}}$ $$\mathbb{G}_{1,3}^{(c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z209}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z210}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z211}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z212}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{5}\left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z213}}$ $\mathbb{G}_{3,2}^{(1,-1;c)}(T_1) = -\frac{\sqrt{5}\left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$

$\boxed{\text{z214}}$ $\mathbb{G}_{3,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{5}\left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{15}$

$\boxed{\text{z215}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z216}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z217}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z218}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(T_1) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(T_1)}{3}$

$\boxed{\text{z219}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(T_1) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{3}$

$\boxed{\text{z220}}$ $\mathbb{G}_{1,3}^{(1,1;c)}(T_1) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{T}_{1,3}^{(b)}(T_1)}{3}$

$\boxed{\text{z398}}$ $\mathbb{Q}_{2,1}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z399}}$ $\mathbb{Q}_{2,2}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z400}}$ $\mathbb{Q}_{2,3}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z401}}$ $\mathbb{Q}_{3,1}^{(c)}(T_2) = -\frac{3\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z402}}$ $\mathbb{Q}_{3,2}^{(c)}(T_2) = \frac{3\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z403}}$ $\mathbb{Q}^{(c)}_{3,3}(T_2) = \frac{\sqrt{3}\mathbb{Q}^{(a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{2,2}(E)}{3}$

$\boxed{\text{z404}}$ $\mathbb{Q}^{(1,-1;c)}_{2,1}(T_2) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,1}(T_1) + \mathbb{M}^{(1,-1;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1) + \mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)\right)}{6}$

$\boxed{\text{z405}}$ $\mathbb{Q}^{(1,-1;c)}_{2,2}(T_2) = \frac{\sqrt{2}\left(-\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,2}(T_1) + \mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1) + \mathbb{M}^{(1,-1;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,2}(T_1) - \mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{6}$

$\boxed{\text{z406}}$ $\mathbb{Q}^{(1,-1;c)}_{2,3}(T_2) = \frac{\sqrt{2}\left(-\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1) - 2\mathbb{M}^{(1,-1;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,3}(T_1) + \mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{6}$

$\boxed{\text{z407}}$ $\mathbb{Q}^{(1,-1;c)}_{3,1}(T_2) = \frac{-\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) + 3\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)}{6}$

$\boxed{\text{z408}}$ $\mathbb{Q}^{(1,-1;c)}_{3,2}(T_2) = -\frac{\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) + 3\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)}{6}$

$\boxed{\text{z409}}$ $\mathbb{Q}^{(1,-1;c)}_{3,3}(T_2) = \frac{\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,1}(E)}{3}$

$\boxed{\text{z410}}$ $\mathbb{Q}^{(1,0;c)}_{2,1}(T_2) = \frac{\sqrt{6}\left(\mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1) + \mathbb{T}^{(1,0;a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1)\right)}{6}$

$\boxed{\text{z411}}$ $\mathbb{Q}^{(1,0;c)}_{2,2}(T_2) = \frac{\sqrt{6}\left(\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1) + \mathbb{T}^{(1,0;a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{6}$

$\boxed{\text{z412}}$ $\mathbb{Q}^{(1,0;c)}_{2,3}(T_2) = \frac{\sqrt{6}\left(\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1) + \mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{6}$

$\boxed{\text{z413}}$ $\mathbb{Q}^{(1,0;c)}_{3,1}(T_2) = -\frac{3\mathbb{Q}^{(1,0;a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,1}(E) + \sqrt{3}\mathbb{Q}^{(1,0;a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,2}(E)}{6}$

$\boxed{\text{z414}}$ $\mathbb{Q}^{(1,0;c)}_{3,2}(T_2) = \frac{3\mathbb{Q}^{(1,0;a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,1}(E) - \sqrt{3}\mathbb{Q}^{(1,0;a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,2}(E)}{6}$

$\boxed{\text{z415}}$ $\mathbb{Q}^{(1,0;c)}_{3,3}(T_2) = \frac{\sqrt{3}\mathbb{Q}^{(1,0;a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{2,2}(E)}{3}$

$\boxed{\text{z416}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(b)}(A_1)}{3}$

$\boxed{\text{z417}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(b)}(A_1)}{3}$

$\boxed{\text{z418}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_0^{(b)}(A_1)}{3}$

$\boxed{\text{z419}}$ $\mathbb{G}_{3,1}^{(1,-1;c)}(T_2) = \frac{-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{6}$

$\boxed{\text{z420}}$ $\mathbb{G}_{3,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) - 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{6}$

$\boxed{\text{z421}}$ $\mathbb{G}_{3,3}^{(1,-1;c)}(T_2) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{3}$

- `A;A_001_1` : 'A'-'A' bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3a@3d`

$\boxed{\text{z11}}$ $\mathbb{Q}_0^{(c)}(A_1, a) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z12}}$ $\mathbb{Q}_0^{(c)}(A_1, b) = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{2}$

$\boxed{\text{z13}}$ $\mathbb{Q}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{2}$

$\boxed{\text{z14}}$ $\mathbb{Q}_0^{(1,1;c)}(A_1) = \mathbb{Q}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z15}}$ $\mathbb{G}_0^{(c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z16}}$ $\mathbb{G}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z17}}$ $$\mathbb{G}_{4}^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z18}}$ $$\mathbb{G}_{0}^{(1,1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z41}}$ $$\mathbb{Q}_{3}^{(1,-1;c)}(A_2) = -\frac{\sqrt{3}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z42}}$ $$\mathbb{Q}_{3}^{(1,0;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z43}}$ $$\mathbb{G}_{3}^{(c)}(A_2) = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{2}$$

$\boxed{\text{z44}}$ $$\mathbb{G}_{3}^{(1,-1;c)}(A_2) = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{2}$$

$\boxed{\text{z83}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

$\boxed{\text{z84}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

$\boxed{\text{z85}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z86}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z87}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

$\boxed{\text{z88}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + \mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

$\boxed{\text{z89}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z90}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,a)=\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z91}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,b)=\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)-\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

$\boxed{\text{z92}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,b)=-\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)+\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

$\boxed{\text{z93}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E)=\frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

$\boxed{\text{z94}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E)=\frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

$\boxed{\text{z95}}$ $$\mathbb{G}_{2,1}^{(c)}(E)=\frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)-\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)+2\mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z96}}$ $$\mathbb{G}_{2,2}^{(c)}(E)=\frac{\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)-\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$$

$\boxed{\text{z97}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,a)=\frac{\sqrt{14}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)-\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)-\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)+\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)+2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{28}$$

$\boxed{\text{z98}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,a)=\frac{\sqrt{14}\left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)-9\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)+2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{84}$$

$\boxed{\text{z99}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,b)=\frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)-\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)+2\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z100}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,b)=\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)-\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$$

$\boxed{\text{z101}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(E)=\frac{\sqrt{14}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)+9\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)-9\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)+2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{84}$$

$\boxed{\text{z102}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(E)=\frac{\sqrt{14}\left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)+\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)-\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)+\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)-2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{28}$$

$\boxed{\text{z103}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$

$\boxed{\text{z104}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E) = \frac{\sqrt{3}\left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z105}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z106}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$

$\boxed{\text{z221}}$ $\mathbb{Q}_{1,1}^{(c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z222}}$ $\mathbb{Q}_{1,2}^{(c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z223}}$ $\mathbb{Q}_{1,3}^{(c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z224}}$ $\mathbb{Q}_{4,1}^{(c)}(T_1) = -\frac{3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z225}}$ $\mathbb{Q}_{4,2}^{(c)}(T_1) = \frac{3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z226}}$ $\mathbb{Q}_{4,3}^{(c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{3}$

$\boxed{\text{z227}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z228}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z229}}$ $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z230}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_1) = \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{12}$$

$\boxed{\text{z231}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_1) = \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$$

$\boxed{\text{z232}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_1) = \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$$

$\boxed{\text{z233}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(T_1) = -\frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$$

$\boxed{\text{z234}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_1) = \frac{3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$$

$\boxed{\text{z235}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{3}$$

$\boxed{\text{z236}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_1) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z237}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_1) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z238}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_1) = \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z239}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_1) = \frac{\sqrt{5}\left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z240}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_1) = -\frac{\sqrt{5}\left(3\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z241}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_1) = \frac{\sqrt{5}\left(3\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{15}$$

$\boxed{\text{z242}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z243}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(T_1) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z244}}$ $$\mathbb{Q}_{1,3}^{(1,1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z245}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_1) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{12}$$

$\boxed{\text{z246}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_1) = \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$$

$\boxed{\text{z247}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$$

$\boxed{\text{z248}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z249}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z250}}$ $$\mathbb{G}_{1,3}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z251}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_1,b) = \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$$

$\boxed{\text{z252}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_1,b) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$$

$\boxed{\text{z253}}$ $$\mathbb{G}_{1,3}^{(1,0;c)}(T_1,b) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)}{3}$$

$\boxed{\text{z422}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_2,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z423}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_2,a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z424}}$ $\mathbb{Q}_{2,3}^{(c)}(T_2, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z425}}$ $\mathbb{Q}_{2,1}^{(c)}(T_2, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z426}}$ $\mathbb{Q}_{2,2}^{(c)}(T_2, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z427}}$ $\mathbb{Q}_{2,3}^{(c)}(T_2, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)}{3}$

$\boxed{\text{z428}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_2, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z429}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_2, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z430}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_2, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z431}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_2, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z432}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_2, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z433}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_2, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)}{3}$

$\boxed{\text{z434}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 4\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$

$\boxed{\text{z435}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + 4\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{12}$

$\boxed{\text{z436}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + 4\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,3}^{(b)}(T_1)}{12}$

$\boxed{\text{z437}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(T_2) = -\frac{3\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$

$\boxed{\text{z438}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_2) = \frac{3\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{6}$$

$\boxed{\text{z439}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{3}$$

$\boxed{\text{z440}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_2) = \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{6}$$

$\boxed{\text{z441}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_2) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{6}$$

$\boxed{\text{z442}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_2) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{3}$$

$\boxed{\text{z443}}$ $$\mathbb{G}_{2,1}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z444}}$ $$\mathbb{G}_{2,2}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z445}}$ $$\mathbb{G}_{2,3}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z446}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{21}$$

$\boxed{\text{z447}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{21}$$

$\boxed{\text{z448}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{21}$$

$\boxed{\text{z449}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z450}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z451}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z452}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 12\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{84}$

$\boxed{\text{z453}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + 12\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{84}$

$\boxed{\text{z454}}$ $\mathbb{G}_{4,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + 12\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{84}$

$\boxed{\text{z455}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(T_2) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z456}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(T_2) = \frac{\sqrt{2}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z457}}$ $\mathbb{G}_{2,3}^{(1,0;c)}(T_2) = \frac{\sqrt{2}\left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - 2\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z458}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z459}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z460}}$ $\mathbb{G}_{2,3}^{(1,1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

- `A;A_002_1` : 'A'-'A' bond-cluster
  - * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  - * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  - * wyckoff: `6b@3c`

$\boxed{\text{z19}}$ $\mathbb{Q}_{0}^{(c)}(A_1) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z20}}$ $\mathbb{G}_{0}^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z45}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z107}}$ $$\mathbb{Q}_{2,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

$\boxed{\text{z108}}$ $$\mathbb{Q}_{2,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

$\boxed{\text{z109}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,a) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z110}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,a) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$$

$\boxed{\text{z111}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)}{2}$$

$\boxed{\text{z112}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,b) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z254}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z255}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_1,a) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z256}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z257}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z258}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z259}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z461}}$ $\mathbb{Q}_{2,1}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(T_2)}{3}$

$\boxed{\text{z462}}$ $\mathbb{Q}_{2,2}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(T_2)}{3}$

$\boxed{\text{z463}}$ $\mathbb{Q}_{2,3}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,3}^{(b)}(T_2)}{3}$

$\boxed{\text{z464}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z465}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z466}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z467}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z468}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

$\boxed{\text{z469}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$

- `A;A_002_1` : '`A`'-'`A`' bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `6b@3c`

$\boxed{\text{z21}}$ $\mathbb{Q}_{0}^{(c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z22}}$ $\mathbb{Q}_{0}^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$

$\boxed{\text{z23}}$ $$\mathbb{Q}_0^{(1,0;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z24}}$ $$\mathbb{G}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{5}\left(\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{5}$$

$\boxed{\text{z25}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_1) = \frac{\sqrt{30}\left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z26}}$ $$\mathbb{G}_0^{(1,1;c)}(A_1) = \mathbb{G}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$$

$\boxed{\text{z46}}$ $$\mathbb{Q}_3^{(c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z47}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_2) = \frac{\sqrt{2}\left(\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{2}$$

$\boxed{\text{z48}}$ $$\mathbb{Q}_3^{(1,0;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z49}}$ $$\mathbb{G}_3^{(c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z50}}$ $$\mathbb{G}_3^{(1,-1;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z51}}$ $$\mathbb{G}_3^{(1,0;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z113}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,a) = \frac{\sqrt{3}\left(-\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z114}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,a) = \frac{\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$$

$\boxed{\text{z115}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,b) = \frac{\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)}{2}$$

$\boxed{\text{z116}}$ $$\mathbb{Q}^{(c)}_{2,2}(E,b) = \frac{\sqrt{3}\left(\mathbb{T}^{(a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2) + \mathbb{T}^{(a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2) - 2\mathbb{T}^{(a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2)\right)}{6}$$

$\boxed{\text{z117}}$ $$\mathbb{Q}^{(1,-1;c)}_{2,1}(E,a) = \frac{-\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1) + \mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)}{2}$$

$\boxed{\text{z118}}$ $$\mathbb{Q}^{(1,-1;c)}_{2,2}(E,a) = \frac{\sqrt{3}\left(-\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1) + 2\mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1)\right)}{6}$$

$\boxed{\text{z119}}$ $$\mathbb{Q}^{(1,-1;c)}_{2,1}(E,b) = \frac{\sqrt{3}\left(\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{M}^{(b)}_{2,1}(T_2) + \mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{M}^{(b)}_{2,2}(T_2) - 2\mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{M}^{(b)}_{2,3}(T_2)\right)}{6}$$

$\boxed{\text{z120}}$ $$\mathbb{Q}^{(1,-1;c)}_{2,2}(E,b) = \frac{-\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{M}^{(b)}_{2,1}(T_2) + \mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{M}^{(b)}_{2,2}(T_2)}{2}$$

$\boxed{\text{z121}}$ $$\mathbb{Q}^{(1,0;c)}_{2,1}(E,a) = \frac{\sqrt{3}\left(-\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1) + 2\mathbb{T}^{(1,0;a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1)\right)}{6}$$

$\boxed{\text{z122}}$ $$\mathbb{Q}^{(1,0;c)}_{2,2}(E,a) = \frac{\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1)}{2}$$

$\boxed{\text{z123}}$ $$\mathbb{Q}^{(1,0;c)}_{2,1}(E,b) = \frac{\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2) - \mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2)}{2}$$

$\boxed{\text{z124}}$ $$\mathbb{Q}^{(1,0;c)}_{2,2}(E,b) = \frac{\sqrt{3}\left(\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2) + \mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2) - 2\mathbb{T}^{(1,0;a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2)\right)}{6}$$

$\boxed{\text{z125}}$ $$\mathbb{G}^{(c)}_{2,1}(E) = \frac{\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,1}(T_2) - \mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,2}(T_2)}{2}$$

$\boxed{\text{z126}}$ $$\mathbb{G}^{(c)}_{2,2}(E) = \frac{\sqrt{3}\left(\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,1}(T_2) + \mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,2}(T_2) - 2\mathbb{Q}^{(a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{2,3}(T_2)\right)}{6}$$

$\boxed{\text{z127}}$ $$\mathbb{G}^{(1,-1;c)}_{2,1}(E,a) = \frac{\sqrt{2}\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{0}(A_1)}{2}$$

$\boxed{\text{z128}}$ $$\mathbb{G}^{(1,-1;c)}_{2,2}(E,a) = \frac{\sqrt{2}\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{0}(A_1)}{2}$$

z129 $\mathbb{G}_{2,1}^{(1,-1;c)}(E,b) = \frac{\sqrt{7}\left(2\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{14}$

z130 $\mathbb{G}_{2,2}^{(1,-1;c)}(E,b) = \frac{\sqrt{7}\left(-2\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{14}$

z131 $\mathbb{G}_{4,1}^{(1,-1;c)}(E) = \frac{\sqrt{21}\left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) + 4\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{42}$

z132 $\mathbb{G}_{4,2}^{(1,-1;c)}(E) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{14}$

z133 $\mathbb{G}_{2,1}^{(1,0;c)}(E) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)}{2}$

z134 $\mathbb{G}_{2,2}^{(1,0;c)}(E) = \frac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) - 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{6}$

z135 $\mathbb{G}_{2,1}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$

z136 $\mathbb{G}_{2,2}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$

z260 $\mathbb{Q}_{1,1}^{(c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

z261 $\mathbb{Q}_{1,2}^{(c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

z262 $\mathbb{Q}_{1,3}^{(c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

z263 $\mathbb{Q}_{1,1}^{(c)}(T_1,b) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 3\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + 3\mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{30}$

z264 $\mathbb{Q}_{1,2}^{(c)}(T_1,b) = \frac{\sqrt{10}\left(3\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 3\mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{30}$

z265 $$\mathbb{Q}^{(c)}_{1,3}(T_1,b)=\frac{\sqrt{10}\left(3\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,2}(T_2)+3\mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,1}(T_2)+2\sqrt{3}\mathbb{Q}^{(a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{2,1}(E)\right)}{30}$$

z266 $$\mathbb{Q}^{(c)}_{3,1}(T_1)=\frac{\sqrt{5}\left(-3\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,1}(E)+3\sqrt{3}\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,2}(E)-2\sqrt{3}\mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,3}(T_2)-2\sqrt{3}\mathbb{Q}^{(a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{2,2}(T_2)\right)}{30}$$

z267 $$\mathbb{Q}^{(c)}_{3,2}(T_1)=-\frac{\sqrt{5}\left(2\sqrt{3}\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,3}(T_2)+3\mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,1}(E)+3\sqrt{3}\mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,2}(E)+2\sqrt{3}\mathbb{Q}^{(a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{2,1}(T_2)\right)}{30}$$

z268 $$\mathbb{Q}^{(c)}_{3,3}(T_1)=\frac{\sqrt{5}\left(-\sqrt{3}\mathbb{Q}^{(a)}_{1,1}(T_1)\mathbb{Q}^{(b)}_{2,2}(T_2)-\sqrt{3}\mathbb{Q}^{(a)}_{1,2}(T_1)\mathbb{Q}^{(b)}_{2,1}(T_2)+3\mathbb{Q}^{(a)}_{1,3}(T_1)\mathbb{Q}^{(b)}_{2,1}(E)\right)}{15}$$

z269 $$\mathbb{Q}^{(1,-1;c)}_{1,1}(T_1)$$
$$=\frac{\sqrt{10}\left(-3\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,1}(T_2)+3\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,1}(E)+\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)-\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,1}(T_2)-\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2)+\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2)\right)}{30}$$

z270 $$\mathbb{Q}^{(1,-1;c)}_{1,2}(T_1)$$
$$=\frac{\sqrt{10}\left(3\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,2}(T_2)+\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2)-\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,2}(T_2)-3\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(E)+\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)-\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2)\right)}{30}$$

z271 $$\mathbb{Q}^{(1,-1;c)}_{1,3}(T_1)=\frac{\sqrt{30}\left(-\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2)+2\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,3}(T_2)+\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2)-2\mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)\right)}{30}$$

z272 $$\mathbb{Q}^{(1,-1;c)}_{3,1}(T_1)$$
$$=\frac{\sqrt{10}\left(3\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,1}(T_2)-3\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,1}(E)-\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)+\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,1}(T_2)-4\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2)+4\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2)\right)}{60}$$

z273 $$\mathbb{Q}^{(1,-1;c)}_{3,2}(T_1)$$
$$=\frac{\sqrt{10}\left(-3\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,2}(T_2)+4\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2)+\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,2}(T_2)+3\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(E)-\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)-4\sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2)\right)}{60}$$

z274 $$\mathbb{Q}^{(1,-1;c)}_{3,3}(T_1)=\frac{\sqrt{30}\left(-2\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2)-\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,3}(T_2)+2\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2)+\mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)\right)}{30}$$

z275 $$\mathbb{Q}^{(1,-1;c)}_{4,1}(T_1)=-\frac{3\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{M}^{(b)}_{2,1}(T_2)+\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,2}(E)\mathbb{M}^{(b)}_{2,1}(T_2)}{6}$$

$\boxed{\text{z276}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_1) = \frac{3\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2)}{6}$$

$\boxed{\text{z277}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{M}_{2,3}^{(b)}(T_2)}{3}$$

$\boxed{\text{z278}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z279}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z280}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z281}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_1,b) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + 3\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z282}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_1,b) = \frac{\sqrt{10}\left(3\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 3\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z283}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_1,b) = \frac{\sqrt{10}\left(3\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{30}$$

$\boxed{\text{z284}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_1) = \frac{\sqrt{5}\left(-3\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) - 2\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) - 2\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z285}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_1) = -\frac{\sqrt{5}\left(2\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z286}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_1) = \frac{\sqrt{5}\left(-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{15}$$

$\boxed{\text{z287}}$ $$\mathbb{G}_{1,1}^{(c)}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z288}}$ $$\mathbb{G}_{1,2}^{(c)}(T_1,a) = \frac{\sqrt{6}\left(-\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z289}}$ $\mathbb{G}^{(c)}_{1,3}(T_1,a)=\frac{\sqrt{6}\left(\mathbb{T}^{(a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1)-\mathbb{T}^{(a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{6}$

$\boxed{\text{z290}}$ $\mathbb{G}^{(c)}_{1,1}(T_1,b)=\frac{\sqrt{6}\left(\mathbb{T}^{(a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2)+\mathbb{T}^{(a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2)\right)}{6}$

$\boxed{\text{z291}}$ $\mathbb{G}^{(c)}_{1,2}(T_1,b)=\frac{\sqrt{6}\left(\mathbb{T}^{(a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2)+\mathbb{T}^{(a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2)\right)}{6}$

$\boxed{\text{z292}}$ $\mathbb{G}^{(c)}_{1,3}(T_1,b)=\frac{\sqrt{6}\left(\mathbb{T}^{(a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2)+\mathbb{T}^{(a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2)\right)}{6}$

$\boxed{\text{z293}}$ $\mathbb{G}^{(1,-1;c)}_{1,1}(T_1,a)=\frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,1}(T_1)+3\mathbb{M}^{(1,-1;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,1}(T_1)+3\mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1)+3\mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)\right)}{30}$

$\boxed{\text{z294}}$ $\mathbb{G}^{(1,-1;c)}_{1,2}(T_1,a)=\frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,2}(T_1)+3\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1)-3\mathbb{M}^{(1,-1;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,2}(T_1)+3\mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{30}$

$\boxed{\text{z295}}$ $\mathbb{G}^{(1,-1;c)}_{1,3}(T_1,a)=\frac{\sqrt{10}\left(2\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,3}(T_1)+3\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)+3\mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{30}$

$\boxed{\text{z296}}$ $\mathbb{G}^{(1,-1;c)}_{1,1}(T_1,b)=\frac{\sqrt{6}\left(-\mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{M}^{(b)}_{2,3}(T_2)+\mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{M}^{(b)}_{2,2}(T_2)\right)}{6}$

$\boxed{\text{z297}}$ $\mathbb{G}^{(1,-1;c)}_{1,2}(T_1,b)=\frac{\sqrt{6}\left(\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{M}^{(b)}_{2,3}(T_2)-\mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{M}^{(b)}_{2,1}(T_2)\right)}{6}$

$\boxed{\text{z298}}$ $\mathbb{G}^{(1,-1;c)}_{1,3}(T_1,b)=\frac{\sqrt{6}\left(-\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{M}^{(b)}_{2,2}(T_2)+\mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{M}^{(b)}_{2,1}(T_2)\right)}{6}$

$\boxed{\text{z299}}$ $\mathbb{G}^{(1,-1;c)}_{3,1}(T_1)=\frac{\sqrt{5}\left(-3\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,1}(T_1)+3\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,1}(T_1)-2\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1)-2\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)\right)}{30}$

$\boxed{\text{z300}}$ $\mathbb{G}^{(1,-1;c)}_{3,2}(T_1)=-\frac{\sqrt{5}\left(3\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,2}(T_1)+2\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1)+3\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,2}(T_1)+2\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{30}$

$\boxed{\text{z301}}$ $\mathbb{G}^{(1,-1;c)}_{3,3}(T_1)=\frac{\sqrt{5}\left(3\mathbb{M}^{(1,-1;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,3}(T_1)-\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)-\sqrt{3}\mathbb{M}^{(1,-1;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{15}$

$$\boxed{\text{z302}}\quad \mathbb{G}^{(1,-1;c)}_{4,1}(T_1) = -\frac{\sqrt{2}\left(3\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,1}(T_2) + 3\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) + \sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(E) + \sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,1}(T_2)\right)}{12}$$

$$\boxed{\text{z303}}\quad \mathbb{G}^{(1,-1;c)}_{4,2}(T_1) = \frac{\sqrt{2}\left(3\mathbb{G}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,2}(T_2) - \sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,2}(T_2) + 3\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) - \sqrt{3}\mathbb{G}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)\right)}{12}$$

$$\boxed{\text{z304}}\quad \mathbb{G}^{(1,-1;c)}_{4,3}(T_1) = \frac{\sqrt{6}\left(\mathbb{G}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,3}(T_2) + \mathbb{G}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)\right)}{6}$$

$$\boxed{\text{z305}}\quad \mathbb{G}^{(1,0;c)}_{1,1}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1) - \mathbb{T}^{(1,0;a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1)\right)}{6}$$

$$\boxed{\text{z306}}\quad \mathbb{G}^{(1,0;c)}_{1,2}(T_1,a) = \frac{\sqrt{6}\left(-\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1) + \mathbb{T}^{(1,0;a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{6}$$

$$\boxed{\text{z307}}\quad \mathbb{G}^{(1,0;c)}_{1,3}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1) - \mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{6}$$

$$\boxed{\text{z308}}\quad \mathbb{G}^{(1,0;c)}_{1,1}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2) + \mathbb{T}^{(1,0;a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2)\right)}{6}$$

$$\boxed{\text{z309}}\quad \mathbb{G}^{(1,0;c)}_{1,2}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2) + \mathbb{T}^{(1,0;a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2)\right)}{6}$$

$$\boxed{\text{z310}}\quad \mathbb{G}^{(1,0;c)}_{1,3}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{T}^{(1,0;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2) + \mathbb{T}^{(1,0;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2)\right)}{6}$$

$$\boxed{\text{z311}}\quad \mathbb{G}^{(1,1;c)}_{1,1}(T_1) = \frac{\sqrt{3}\mathbb{M}^{(1,1;a)}_{0}(A_1)\mathbb{T}^{(b)}_{1,1}(T_1)}{3}$$

$$\boxed{\text{z312}}\quad \mathbb{G}^{(1,1;c)}_{1,2}(T_1) = \frac{\sqrt{3}\mathbb{M}^{(1,1;a)}_{0}(A_1)\mathbb{T}^{(b)}_{1,2}(T_1)}{3}$$

$$\boxed{\text{z313}}\quad \mathbb{G}^{(1,1;c)}_{1,3}(T_1) = \frac{\sqrt{3}\mathbb{M}^{(1,1;a)}_{0}(A_1)\mathbb{T}^{(b)}_{1,3}(T_1)}{3}$$

$$\boxed{\text{z470}}\quad \mathbb{Q}^{(c)}_{2,1}(T_2,a) = \frac{\sqrt{6}\left(\mathbb{T}^{(a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1) + \mathbb{T}^{(a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1)\right)}{6}$$

$$\boxed{\text{z471}}\quad \mathbb{Q}_{2,2}^{(c)}(T_2,a) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z472}}\quad \mathbb{Q}_{2,3}^{(c)}(T_2,a) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z473}}\quad \mathbb{Q}_{2,1}^{(c)}(T_2,b) = \frac{\sqrt{6}\left(-\mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z474}}\quad \mathbb{Q}_{2,2}^{(c)}(T_2,b) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) - \mathbb{T}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z475}}\quad \mathbb{Q}_{2,3}^{(c)}(T_2,b) = \frac{\sqrt{6}\left(-\mathbb{T}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{T}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z476}}\quad \mathbb{Q}_{3,1}^{(c)}(T_2) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 2\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) - 2\mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)}{6}$$

$$\boxed{\text{z477}}\quad \mathbb{Q}_{3,2}^{(c)}(T_2) = \frac{-2\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 2\mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)}{6}$$

$$\boxed{\text{z478}}\quad \mathbb{Q}_{3,3}^{(c)}(T_2) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{3}$$

$$\boxed{\text{z479}}\quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z480}}\quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{2}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z481}}\quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{2}\left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z482}}\quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{10}\left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{M}_{2,1}^{(b)}(T_2) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{M}_{2,1}^{(b)}(T_2) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{30}$$

$$\boxed{\text{z483}}\quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{10}\left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{30}$$

$$\boxed{\text{z484}}\quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{10}\left(-2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{M}_{2,3}^{(b)}(T_2) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{30}$$

$$\boxed{\text{z485}}\quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{12}$$

$$\boxed{\text{z486}}\quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{12}$$

$$\boxed{\text{z487}}\quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{6}$$

$$\boxed{\text{z488}}\quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{5}\left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{M}_{2,1}^{(b)}(T_2) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{M}_{2,1}^{(b)}(T_2) + 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{30}$$

$$\boxed{\text{z489}}\quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{5}\left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) - 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{30}$$

$$\boxed{\text{z490}}\quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{5}\left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{M}_{2,3}^{(b)}(T_2) + \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{15}$$

$$\boxed{\text{z491}}\quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_2,a) = \frac{\sqrt{6}\left(\mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z492}}\quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_2,a) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z493}}\quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_2,a) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z494}}\quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_2,b) = \frac{\sqrt{6}\left(-\mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z495}}\quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_2,b) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) - \mathbb{T}_{1,3}^{(1,0;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z496}}\quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_2,b) = \frac{\sqrt{6}\left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{T}_{1,2}^{(1,0;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z497}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_2) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 2\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) - 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)}{6}$$

$\boxed{\text{z498}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_2) = \frac{-2\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)}{6}$$

$\boxed{\text{z499}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_2) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{3}$$

$\boxed{\text{z500}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(T_2) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{M}_{2,1}^{(b)}(T_2)}{3}$$

$\boxed{\text{z501}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(T_2) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{M}_{2,2}^{(b)}(T_2)}{3}$$

$\boxed{\text{z502}}$ $$\mathbb{Q}_{2,3}^{(1,1;c)}(T_2) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{M}_{2,3}^{(b)}(T_2)}{3}$$

$\boxed{\text{z503}}$ $$\mathbb{G}_{2,1}^{(c)}(T_2) = \frac{\sqrt{2}\left(-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + \mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z504}}$ $$\mathbb{G}_{2,2}^{(c)}(T_2) = \frac{\sqrt{2}\left(\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z505}}$ $$\mathbb{G}_{2,3}^{(c)}(T_2) = \frac{\sqrt{2}\left(-\mathbb{Q}_{1,1}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) + \mathbb{Q}_{1,2}^{(a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + 2\mathbb{Q}_{1,3}^{(a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{6}$$

$\boxed{\text{z506}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_2, a) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z507}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_2, a) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z508}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_2, a) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z509}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_2, b)$$
$$= \frac{\sqrt{14}\left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2) + 3\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{42}$$

$\boxed{\text{z510}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_2,b)$

$$= \frac{\sqrt{14}\left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)+3\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)+3\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)+\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)+3\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)+3\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{42}$$

$\boxed{\text{z511}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{14}\left(-2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2)+3\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)+3\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)-2\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{42}$

$\boxed{\text{z512}}$ $\mathbb{G}_{3,1}^{(1,-1;c)}(T_2) = \frac{-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1)-\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1)-2\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)+2\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{6}$

$\boxed{\text{z513}}$ $\mathbb{G}_{3,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1)+2\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)-\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1)-2\mathbb{M}_{2,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{6}$

$\boxed{\text{z514}}$ $\mathbb{G}_{3,3}^{(1,-1;c)}(T_2) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)+\mathbb{M}_{2,2}^{(1,-1;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1)+\mathbb{M}_{2,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{3}$

$\boxed{\text{z515}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(T_2)$

$$= \frac{\sqrt{14}\left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)-3\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)+3\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)+3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)+4\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)+4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{84}$$

$\boxed{\text{z516}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(T_2)$

$$= \frac{\sqrt{14}\left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)+4\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)-3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)-3\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)-3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)+4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{84}$$

$\boxed{\text{z517}}$ $\mathbb{G}_{4,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{14}\left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2)+2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)+2\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)+3\mathbb{G}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{42}$

$\boxed{\text{z518}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(T_2) = \frac{\sqrt{2}\left(-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)-\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)-\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)+\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z519}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(T_2) = \frac{\sqrt{2}\left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)+\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)-\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)-\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z520}}$ $\mathbb{G}_{2,3}^{(1,0;c)}(T_2) = \frac{\sqrt{2}\left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)+\mathbb{Q}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)+2\mathbb{Q}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{6}$

$\boxed{\text{z521}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(T_2)}{3}$

$\boxed{\text{z522}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(T_2)}{3}$

$\boxed{\text{z523}}$ $\mathbb{G}_{2,3}^{(1,1;c)}(T_2) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,3}^{(b)}(T_2)}{3}$

- `A;A_002_1` : '`A`'-'`A`' bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `6b@3c`

$\boxed{\text{z27}}$ $\mathbb{Q}_{0}^{(c)}(A_1,a) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z28}}$ $\mathbb{Q}_{0}^{(c)}(A_1,b) = \frac{\sqrt{5}\left(\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + \mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) + \mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{5}$

$\boxed{\text{z29}}$ $\mathbb{Q}_{4}^{(c)}(A_1) = \frac{\sqrt{30}\left(3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - 2\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) - 2\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{30}$

$\boxed{\text{z30}}$ $\mathbb{Q}_{0}^{(1,-1;c)}(A_1) = \frac{\sqrt{5}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{5}$

$\boxed{\text{z31}}$ $\mathbb{Q}_{4}^{(1,-1;c)}(A_1) = \frac{\sqrt{30}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{30}$

$\boxed{\text{z32}}$ $\mathbb{Q}_{0}^{(1,1;c)}(A_1) = \mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z33}}$ $\mathbb{G}_{0}^{(c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z34}}$ $\mathbb{G}_{0}^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z35}}$ $\mathbb{G}_{4}^{(1,-1;c)}(A_1,a) = \frac{\sqrt{3}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$

$\boxed{\text{z36}}$ $\mathbb{G}_{4}^{(1,-1;c)}(A_1,b) = -\frac{\sqrt{3}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$

$\boxed{\text{z37}}$ $$\mathbb{G}_0^{(1,0;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z38}}$ $$\mathbb{G}_0^{(1,1;c)}(A_1) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z52}}$ $$\mathbb{Q}_3^{(c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z53}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_2,a) = -\frac{\sqrt{3}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z54}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_2,b) = -\frac{\sqrt{3}\left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z55}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_2,c) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z56}}$ $$\mathbb{Q}_3^{(1,0;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{3}$$

$\boxed{\text{z57}}$ $$\mathbb{Q}_3^{(1,1;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z58}}$ $$\mathbb{G}_3^{(c)}(A_2) = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{2}$$

$\boxed{\text{z59}}$ $$\mathbb{G}_3^{(1,-1;c)}(A_2) = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{2}$$

$\boxed{\text{z60}}$ $$\mathbb{G}_3^{(1,0;c)}(A_2) = \frac{\sqrt{3}\left(\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) + \mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{3}$$

$\boxed{\text{z137}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

$\boxed{\text{z138}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

$\boxed{\text{z139}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z140}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z141}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,c) = \frac{\sqrt{7}\left(2\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 2\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + \mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) - 2\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{14}$$

$\boxed{\text{z142}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,c) = \frac{\sqrt{7}\left(-2\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 2\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{14}$$

$\boxed{\text{z143}}$ $$\mathbb{Q}_{4,1}^{(c)}(E) = \frac{\sqrt{21}\left(3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 3\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - 2\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) + 4\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{42}$$

$\boxed{\text{z144}}$ $$\mathbb{Q}_{4,2}^{(c)}(E) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + 2\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{14}$$

$\boxed{\text{z145}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z146}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z147}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,b) = \frac{\sqrt{7}\left(2\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{14}$$

$\boxed{\text{z148}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,b) = \frac{\sqrt{7}\left(-2\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{14}$$

$\boxed{\text{z149}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E) = \frac{\sqrt{21}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) + 4\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{42}$$

$\boxed{\text{z150}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{14}$$

$\boxed{\text{z151}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(E) = \frac{\sqrt{14}\left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{28}$$

$$\boxed{\text{z152}}\quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E) = \frac{\sqrt{14}\left(\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) - 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{84}$$

$$\boxed{\text{z153}}\quad \mathbb{Q}_{2,1}^{(1,0;c)}(E) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) - \mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)}{2}$$

$$\boxed{\text{z154}}\quad \mathbb{Q}_{2,2}^{(1,0;c)}(E) = \frac{\sqrt{3}\left(\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + \mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) - 2\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z155}}\quad \mathbb{Q}_{2,1}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z156}}\quad \mathbb{Q}_{2,2}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E)}{2}$$

$$\boxed{\text{z157}}\quad \mathbb{G}_{2,1}^{(c)}(E,a) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z158}}\quad \mathbb{G}_{2,2}^{(c)}(E,a) = \frac{\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)}{2}$$

$$\boxed{\text{z159}}\quad \mathbb{G}_{2,1}^{(c)}(E,b) = \frac{\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)}{2}$$

$$\boxed{\text{z160}}\quad \mathbb{G}_{2,2}^{(c)}(E,b) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - 2\mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z161}}\quad \mathbb{G}_{2,1}^{(1,-1;c)}(E,a) = \frac{\sqrt{14}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{28}$$

$$\boxed{\text{z162}}\quad \mathbb{G}_{2,2}^{(1,-1;c)}(E,a) = \frac{\sqrt{14}\left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{84}$$

$$\boxed{\text{z163}}\quad \mathbb{G}_{2,1}^{(1,-1;c)}(E,b) = \frac{\sqrt{14}\left(-9\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) + 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{84}$$

$$\boxed{\text{z164}}\quad \mathbb{G}_{2,2}^{(1,-1;c)}(E,b) = \frac{\sqrt{14}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{28}$$

$$\boxed{\text{z165}}\quad \mathbb{G}^{(1,-1;c)}_{2,1}(E,c) = \frac{\sqrt{3}\left(-\mathbb{M}^{(1,-1;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{M}^{(1,-1;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1) + 2\mathbb{M}^{(1,-1;a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1)\right)}{6}$$

$$\boxed{\text{z166}}\quad \mathbb{G}^{(1,-1;c)}_{2,2}(E,c) = \frac{\mathbb{M}^{(1,-1;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{M}^{(1,-1;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1)}{2}$$

$$\boxed{\text{z167}}\quad \mathbb{G}^{(1,-1;c)}_{2,1}(E,d) = \frac{\mathbb{M}^{(1,-1;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2) - \mathbb{M}^{(1,-1;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2)}{2}$$

$$\boxed{\text{z168}}\quad \mathbb{G}^{(1,-1;c)}_{2,2}(E,d) = \frac{\sqrt{3}\left(\mathbb{M}^{(1,-1;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2) + \mathbb{M}^{(1,-1;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2) - 2\mathbb{M}^{(1,-1;a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2)\right)}{6}$$

$$\boxed{\text{z169}}\quad \mathbb{G}^{(1,-1;c)}_{4,1}(E) = \frac{\sqrt{14}\left(-\sqrt{15}\mathbb{M}^{(1,-1;a)}_{3,1}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) + 9\mathbb{M}^{(1,-1;a)}_{3,1}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1) - \sqrt{15}\mathbb{M}^{(1,-1;a)}_{3,2}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1) - 9\mathbb{M}^{(1,-1;a)}_{3,2}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1) + 2\sqrt{15}\mathbb{M}^{(1,-1;a)}_{3,3}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1)\right)}{84}$$

$$\boxed{\text{z170}}\quad \mathbb{G}^{(1,-1;c)}_{4,2}(E) = \frac{\sqrt{14}\left(\sqrt{5}\mathbb{M}^{(1,-1;a)}_{3,1}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) + \sqrt{3}\mathbb{M}^{(1,-1;a)}_{3,1}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1) - \sqrt{5}\mathbb{M}^{(1,-1;a)}_{3,2}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1) + \sqrt{3}\mathbb{M}^{(1,-1;a)}_{3,2}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1) - 2\sqrt{3}\mathbb{M}^{(1,-1;a)}_{3,3}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1)\right)}{28}$$

$$\boxed{\text{z171}}\quad \mathbb{G}^{(1,0;c)}_{2,1}(E,a) = \frac{-\mathbb{T}^{(1,0;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1) + \mathbb{T}^{(1,0;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)}{2}$$

$$\boxed{\text{z172}}\quad \mathbb{G}^{(1,0;c)}_{2,2}(E,a) = \frac{\sqrt{3}\left(-\mathbb{T}^{(1,0;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{T}^{(1,0;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1) + 2\mathbb{T}^{(1,0;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1)\right)}{6}$$

$$\boxed{\text{z173}}\quad \mathbb{G}^{(1,0;c)}_{2,1}(E,b) = \frac{\sqrt{3}\left(\mathbb{T}^{(1,0;a)}_{2,1}(T_2)\mathbb{M}^{(b)}_{2,1}(T_2) + \mathbb{T}^{(1,0;a)}_{2,2}(T_2)\mathbb{M}^{(b)}_{2,2}(T_2) - 2\mathbb{T}^{(1,0;a)}_{2,3}(T_2)\mathbb{M}^{(b)}_{2,3}(T_2)\right)}{6}$$

$$\boxed{\text{z174}}\quad \mathbb{G}^{(1,0;c)}_{2,2}(E,b) = \frac{-\mathbb{T}^{(1,0;a)}_{2,1}(T_2)\mathbb{M}^{(b)}_{2,1}(T_2) + \mathbb{T}^{(1,0;a)}_{2,2}(T_2)\mathbb{M}^{(b)}_{2,2}(T_2)}{2}$$

$$\boxed{\text{z175}}\quad \mathbb{G}^{(1,1;c)}_{2,1}(E,a) = \frac{\sqrt{3}\left(-\mathbb{M}^{(1,1;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{M}^{(1,1;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1) + 2\mathbb{M}^{(1,1;a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1)\right)}{6}$$

$$\boxed{\text{z176}}\quad \mathbb{G}^{(1,1;c)}_{2,2}(E,a) = \frac{\mathbb{M}^{(1,1;a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) - \mathbb{M}^{(1,1;a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1)}{2}$$

$$\boxed{\text{z177}}\quad \mathbb{G}^{(1,1;c)}_{2,1}(E,b) = \frac{\mathbb{M}^{(1,1;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2) - \mathbb{M}^{(1,1;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2)}{2}$$

$\boxed{\text{z178}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E,b) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - 2\mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z314}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z315}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_1,a) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z316}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z317}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z318}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z319}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z320}}$ $$\mathbb{Q}_{4,1}^{(c)}(T_1) = -\frac{\sqrt{2}\left(3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{12}$$

$\boxed{\text{z321}}$ $$\mathbb{Q}_{4,2}^{(c)}(T_1) = \frac{\sqrt{2}\left(3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) + 3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{12}$$

$\boxed{\text{z322}}$ $$\mathbb{Q}_{4,3}^{(c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2) + \mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{6}$$

$\boxed{\text{z323}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_1,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{21}$$

$\boxed{\text{z324}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_1,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{21}$$

$\boxed{\text{z325}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_1,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{21}$$

$$\boxed{\text{z326}}\quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z327}}\quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z328}}\quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z329}}\quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_1,c) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z330}}\quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_1,c) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z331}}\quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_1,c) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z332}}\quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_1,a) = \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{12}$$

$$\boxed{\text{z333}}\quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_1,a) = \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$$

$$\boxed{\text{z334}}\quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_1,a) = \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$$

$$\boxed{\text{z335}}\quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) - 5\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + 5\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) - 16\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{84}$$

$$\boxed{\text{z336}}\quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + 5\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - 5\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) - 16\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{84}$$

$$\boxed{\text{z337}}\quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_1,b) = \frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - 5\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + 5\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2) - 16\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{M}_{2,3}^{(b)}(T_2)\right)}{84}$$

$$\boxed{\text{z338}}\quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_1) = -\frac{\sqrt{2}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{12}$$

$\boxed{\text{z339}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_1) = \frac{\sqrt{2}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{12}$$

$\boxed{\text{z340}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{6}\left(\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{6}$$

$\boxed{\text{z341}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(T_1,2) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)}{12}$$

$\boxed{\text{z342}}$ $$\mathbb{Q}_{5,2}^{(1,-1;c)}(T_1,2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)}{12}$$

$\boxed{\text{z343}}$ $$\mathbb{Q}_{5,3}^{(1,-1;c)}(T_1,2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)}{12}$$

$\boxed{\text{z344}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_1,a) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z345}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_1,a) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z346}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_1,a) = \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z347}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_1,b) = \frac{\sqrt{2}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{M}_{2,1}^{(b)}(T_2) - \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{M}_{2,1}^{(b)}(T_2) - \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z348}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_1,b) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) - \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) - \mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z349}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_1,b) = \frac{\sqrt{2}\left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + 2\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z350}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_1,a) = \frac{\sqrt{5}\left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{30}$$

$\boxed{\text{z351}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_1,a) = -\frac{\sqrt{5}\left(3\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{30}$$

$$\boxed{\text{z352}}\quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_1,a) = \frac{\sqrt{5}\left(3\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{15}$$

$$\boxed{\text{z353}}\quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_1,b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{M}_{2,1}^{(b)}(T_2) + \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{M}_{2,1}^{(b)}(T_2) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)}{6}$$

$$\boxed{\text{z354}}\quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_1,b) = \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{M}_{2,2}^{(b)}(T_2) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)}{6}$$

$$\boxed{\text{z355}}\quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_1,b) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) - \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)}{3}$$

$$\boxed{\text{z356}}\quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z357}}\quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_1,a) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z358}}\quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_1,a) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z359}}\quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z360}}\quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z361}}\quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_1,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) + \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z362}}\quad \mathbb{G}_{1,1}^{(c)}(T_1) = \frac{\sqrt{10}\left(-3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2) + \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{30}$$

$$\boxed{\text{z363}}\quad \mathbb{G}_{1,2}^{(c)}(T_1) = \frac{\sqrt{10}\left(3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) - 3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) - \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{30}$$

$$\boxed{\text{z364}}\quad \mathbb{G}_{1,3}^{(c)}(T_1) = \frac{\sqrt{30}\left(-\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) + 2\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2) + \mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) - 2\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{30}$$

$\boxed{\text{z365}}$ $$\mathbb{G}^{(c)}_{3,1}(T_1) = \frac{\sqrt{10}\left(3\mathbb{Q}^{(a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,1}(T_2) - 3\mathbb{Q}^{(a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) - \sqrt{3}\mathbb{Q}^{(a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(E) + \sqrt{3}\mathbb{Q}^{(a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,1}(T_2) - 4\sqrt{3}\mathbb{Q}^{(a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2) + 4\sqrt{3}\mathbb{Q}^{(a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2)\right)}{60}$$

$\boxed{\text{z366}}$ $$\mathbb{G}^{(c)}_{3,2}(T_1) = \frac{\sqrt{10}\left(-3\mathbb{Q}^{(a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,2}(T_2) + 4\sqrt{3}\mathbb{Q}^{(a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2) + \sqrt{3}\mathbb{Q}^{(a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,2}(T_2) + 3\mathbb{Q}^{(a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) - \sqrt{3}\mathbb{Q}^{(a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,2}(E) - 4\sqrt{3}\mathbb{Q}^{(a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2)\right)}{60}$$

$\boxed{\text{z367}}$ $$\mathbb{G}^{(c)}_{3,3}(T_1) = \frac{\sqrt{30}\left(-2\mathbb{Q}^{(a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2) - \mathbb{Q}^{(a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,3}(T_2) + 2\mathbb{Q}^{(a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2) + \mathbb{Q}^{(a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)\right)}{30}$$

$\boxed{\text{z368}}$ $\mathbb{G}^{(1,-1;c)}_{1,1}(T_1)$
$$= \frac{\sqrt{10}\left(-3\mathbb{Q}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,1}(T_2) + 3\mathbb{Q}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) + \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(E) - \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,1}(T_2) - \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2) + \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2)\right)}{30}$$

$\boxed{\text{z369}}$ $\mathbb{G}^{(1,-1;c)}_{1,2}(T_1)$
$$= \frac{\sqrt{10}\left(3\mathbb{Q}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,2}(T_2) + \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2) - \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,2}(T_2) - 3\mathbb{Q}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) + \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,2}(E) - \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2)\right)}{30}$$

$\boxed{\text{z370}}$ $$\mathbb{G}^{(1,-1;c)}_{1,3}(T_1) = \frac{\sqrt{30}\left(-\mathbb{Q}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2) + 2\mathbb{Q}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,3}(T_2) + \mathbb{Q}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2) - 2\mathbb{Q}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)\right)}{30}$$

$\boxed{\text{z371}}$ $\mathbb{G}^{(1,-1;c)}_{3,1}(T_1)$
$$= \frac{\sqrt{10}\left(3\mathbb{Q}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,1}(T_2) - 3\mathbb{Q}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) - \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(E) + \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,1}(T_2) - 4\sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2) + 4\sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2)\right)}{60}$$

$\boxed{\text{z372}}$ $\mathbb{G}^{(1,-1;c)}_{3,2}(T_1)$
$$= \frac{\sqrt{10}\left(-3\mathbb{Q}^{(1,-1;a)}_{2,1}(E)\mathbb{Q}^{(b)}_{2,2}(T_2) + 4\sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,3}(T_2) + \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,2}(T_2) + 3\mathbb{Q}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(E) - \sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,2}(E) - 4\sqrt{3}\mathbb{Q}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2)\right)}{60}$$

$\boxed{\text{z373}}$ $$\mathbb{G}^{(1,-1;c)}_{3,3}(T_1) = \frac{\sqrt{30}\left(-2\mathbb{Q}^{(1,-1;a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{2,2}(T_2) - \mathbb{Q}^{(1,-1;a)}_{2,2}(E)\mathbb{Q}^{(b)}_{2,3}(T_2) + 2\mathbb{Q}^{(1,-1;a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{2,1}(T_2) + \mathbb{Q}^{(1,-1;a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{2,2}(E)\right)}{30}$$

$\boxed{\text{z374}}$ $$\mathbb{G}^{(1,-1;c)}_{4,1}(T_1) = \frac{\sqrt{15}\mathbb{M}^{(1,-1;a)}_{3,2}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1) - 3\mathbb{M}^{(1,-1;a)}_{3,2}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1) - \sqrt{15}\mathbb{M}^{(1,-1;a)}_{3,3}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1) - 3\mathbb{M}^{(1,-1;a)}_{3,3}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)}{12}$$

$\boxed{\text{z375}}$ $$\mathbb{G}^{(1,-1;c)}_{4,2}(T_1) = \frac{-\sqrt{15}\mathbb{M}^{(1,-1;a)}_{3,1}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1) - 3\mathbb{M}^{(1,-1;a)}_{3,1}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1) + \sqrt{15}\mathbb{M}^{(1,-1;a)}_{3,3}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1) - 3\mathbb{M}^{(1,-1;a)}_{3,3}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)}{12}$$

$\boxed{\text{z376}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_1) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$$

$\boxed{\text{z377}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_1, a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z378}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_1, a) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z379}}$ $$\mathbb{G}_{1,3}^{(1,0;c)}(T_1, a) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z380}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_1, b) = \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z381}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_1, b) = \frac{\sqrt{10}\left(3\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z382}}$ $$\mathbb{G}_{1,3}^{(1,0;c)}(T_1, b) = \frac{\sqrt{10}\left(3\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{30}$$

$\boxed{\text{z383}}$ $$\mathbb{G}_{3,1}^{(1,0;c)}(T_1) = \frac{\sqrt{5}\left(-3\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) - 2\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) - 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z384}}$ $$\mathbb{G}_{3,2}^{(1,0;c)}(T_1) = -\frac{\sqrt{5}\left(2\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) + 3\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) + 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{30}$$

$\boxed{\text{z385}}$ $$\mathbb{G}_{3,3}^{(1,0;c)}(T_1) = \frac{\sqrt{5}\left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{15}$$

$\boxed{\text{z524}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_2, a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(T_2)}{3}$$

$\boxed{\text{z525}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_2, a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(T_2)}{3}$$

$\boxed{\text{z526}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_2, a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,3}^{(b)}(T_2)}{3}$$

$\boxed{\text{z527}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_2,b)=\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z528}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_2,b)=\frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z529}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_2,b)=\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z530}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_2,c)=\frac{\sqrt{14}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)+\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)-3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)-3\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)+3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)+3\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{42}$$

$\boxed{\text{z531}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_2,c)=\frac{\sqrt{14}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)+3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)+3\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)+\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)+3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)+3\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{42}$$

$\boxed{\text{z532}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_2,c)=\frac{\sqrt{14}\left(-2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2)+3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)+3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)-2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{42}$$

$\boxed{\text{z533}}$ $$\mathbb{Q}_{3,1}^{(c)}(T_2)=\frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)-\mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z534}}$ $$\mathbb{Q}_{3,2}^{(c)}(T_2)=\frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)+\mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z535}}$ $$\mathbb{Q}_{3,3}^{(c)}(T_2)=\frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)-\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$\boxed{\text{z536}}$ $$\mathbb{Q}_{4,1}^{(c)}(T_2)=\frac{\sqrt{14}\left(-3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)-3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)+3\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)+3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)+4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)+4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{84}$$

$\boxed{\text{z537}}$ $$\mathbb{Q}_{4,2}^{(c)}(T_2)=\frac{\sqrt{14}\left(-3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)+4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)-3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)-3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)-3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)+4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{84}$$

$\boxed{\text{z538}}$ $$\mathbb{Q}_{4,3}^{(c)}(T_2)=\frac{\sqrt{14}\left(3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2)+2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)+2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)+3\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{42}$$

$\boxed{\text{z539}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_2,a)=\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$$

$\boxed{\text{z540}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_2,a)=\frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z541}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_2,a)=\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z542}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_2,b)$
$=\frac{\sqrt{14}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)+\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)-3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)-3\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)+3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)+3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{42}$

$\boxed{\text{z543}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_2,b)$
$=\frac{\sqrt{14}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)+3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2)+3\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2)+\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)+3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)+3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{42}$

$\boxed{\text{z544}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_2,b)=\frac{\sqrt{14}\left(-2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2)+3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)+3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)-2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{42}$

$\boxed{\text{z545}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_2,a)=\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)+\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)+\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)-\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)+4\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(T_1)}{12}$

$\boxed{\text{z546}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_2,a)=\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)-\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)+\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)+\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)+4\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{12}$

$\boxed{\text{z547}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_2,a)=\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)+\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)+\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)-\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)+4\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,3}^{(b)}(T_1)}{12}$

$\boxed{\text{z548}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_2,b)=\frac{\sqrt{2}\left(\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)-3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)-\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)-3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{24}$

$\boxed{\text{z549}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_2,b)=\frac{\sqrt{2}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)-3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)+\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)-3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{24}$

$\boxed{\text{z550}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_2,b)=\frac{\sqrt{2}\left(\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)-3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)-\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)-3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{24}$

$\boxed{\text{z551}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_2,c)=\frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2)-\mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z552}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_2,c) = \dfrac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z553}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_2,c) = \dfrac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z554}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(T_2)$

$$= \frac{\sqrt{14}\left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + 3\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{84}$$

$\boxed{\text{z555}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(T_2)$

$$= \frac{\sqrt{14}\left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,3}^{(b)}(T_2) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{84}$$

$\boxed{\text{z556}}$ $\mathbb{Q}_{4,3}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{14}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(T_2) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(T_2) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{42}$

$\boxed{\text{z557}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{2}\left(-3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) + 3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{24}$

$\boxed{\text{z558}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{2}\left(3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,3}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2) - 3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{24}$

$\boxed{\text{z559}}$ $\mathbb{Q}_{5,3}^{(1,-1;c)}(T_2) = \dfrac{\sqrt{2}\left(-3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{M}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2) + 3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{M}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{24}$

$\boxed{\text{z560}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(T_2) = \dfrac{\sqrt{2}\left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + \mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z561}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(T_2) = \dfrac{\sqrt{2}\left(\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2) + \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E) - \mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{6}$

$\boxed{\text{z562}}$ $\mathbb{Q}_{2,3}^{(1,0;c)}(T_2) = \dfrac{\sqrt{2}\left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2) + \mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{6}$

$\boxed{\text{z563}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(T_2,a) = \dfrac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - \mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)}{6}$

$$\boxed{\text{z564}}\quad \mathbb{Q}^{(1,0;c)}_{3,2}(T_2,a)=\frac{\sqrt{3}\mathbb{T}^{(1,0;a)}_{2,1}(E)\mathbb{T}^{(b)}_{1,2}(T_1)+2\mathbb{T}^{(1,0;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,3}(T_1)-\mathbb{T}^{(1,0;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,2}(T_1)-2\mathbb{T}^{(1,0;a)}_{2,3}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)}{6}$$

$$\boxed{\text{z565}}\quad \mathbb{Q}^{(1,0;c)}_{3,3}(T_2,a)=\frac{-\mathbb{T}^{(1,0;a)}_{2,1}(T_2)\mathbb{T}^{(b)}_{1,2}(T_1)+\mathbb{T}^{(1,0;a)}_{2,2}(E)\mathbb{T}^{(b)}_{1,3}(T_1)+\mathbb{T}^{(1,0;a)}_{2,2}(T_2)\mathbb{T}^{(b)}_{1,1}(T_1)}{3}$$

$$\boxed{\text{z566}}\quad \mathbb{Q}^{(1,0;c)}_{3,1}(T_2,b)=\frac{\sqrt{3}\mathbb{T}^{(1,0;a)}_{2,1}(E)\mathbb{M}^{(b)}_{2,1}(T_2)-3\mathbb{T}^{(1,0;a)}_{2,2}(E)\mathbb{M}^{(b)}_{2,1}(T_2)}{6}$$

$$\boxed{\text{z567}}\quad \mathbb{Q}^{(1,0;c)}_{3,2}(T_2,b)=\frac{\sqrt{3}\mathbb{T}^{(1,0;a)}_{2,1}(E)\mathbb{M}^{(b)}_{2,2}(T_2)+3\mathbb{T}^{(1,0;a)}_{2,2}(E)\mathbb{M}^{(b)}_{2,2}(T_2)}{6}$$

$$\boxed{\text{z568}}\quad \mathbb{Q}^{(1,0;c)}_{3,3}(T_2,b)=-\frac{\sqrt{3}\mathbb{T}^{(1,0;a)}_{2,1}(E)\mathbb{M}^{(b)}_{2,3}(T_2)}{3}$$

$$\boxed{\text{z569}}\quad \mathbb{Q}^{(1,1;c)}_{2,1}(T_2)=\frac{\sqrt{3}\mathbb{Q}^{(1,1;a)}_{0}(A_1)\mathbb{Q}^{(b)}_{2,1}(T_2)}{3}$$

$$\boxed{\text{z570}}\quad \mathbb{Q}^{(1,1;c)}_{2,2}(T_2)=\frac{\sqrt{3}\mathbb{Q}^{(1,1;a)}_{0}(A_1)\mathbb{Q}^{(b)}_{2,2}(T_2)}{3}$$

$$\boxed{\text{z571}}\quad \mathbb{Q}^{(1,1;c)}_{2,3}(T_2)=\frac{\sqrt{3}\mathbb{Q}^{(1,1;a)}_{0}(A_1)\mathbb{Q}^{(b)}_{2,3}(T_2)}{3}$$

$$\boxed{\text{z572}}\quad \mathbb{Q}^{(1,1;c)}_{3,1}(T_2)=\frac{\sqrt{6}\left(\mathbb{M}^{(1,1;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2)-\mathbb{M}^{(1,1;a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2)\right)}{6}$$

$$\boxed{\text{z573}}\quad \mathbb{Q}^{(1,1;c)}_{3,2}(T_2)=\frac{\sqrt{6}\left(-\mathbb{M}^{(1,1;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,3}(T_2)+\mathbb{M}^{(1,1;a)}_{1,3}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2)\right)}{6}$$

$$\boxed{\text{z574}}\quad \mathbb{Q}^{(1,1;c)}_{3,3}(T_2)=\frac{\sqrt{6}\left(\mathbb{M}^{(1,1;a)}_{1,1}(T_1)\mathbb{M}^{(b)}_{2,2}(T_2)-\mathbb{M}^{(1,1;a)}_{1,2}(T_1)\mathbb{M}^{(b)}_{2,1}(T_2)\right)}{6}$$

$$\boxed{\text{z575}}\quad \mathbb{G}^{(c)}_{2,1}(T_2)=\frac{\sqrt{6}\left(\mathbb{M}^{(a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1)+\mathbb{M}^{(a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,2}(T_1)\right)}{6}$$

$$\boxed{\text{z576}}\quad \mathbb{G}^{(c)}_{2,2}(T_2)=\frac{\sqrt{6}\left(\mathbb{M}^{(a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,3}(T_1)+\mathbb{M}^{(a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,1}(T_1)\right)}{6}$$

$\boxed{\text{z577}}$ $$\mathbb{G}_{2,3}^{(c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z578}}$ $$\mathbb{G}_{3,1}^{(c)}(T_2) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) - 3\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{12}$$

$\boxed{\text{z579}}$ $$\mathbb{G}_{3,2}^{(c)}(T_2) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) + 3\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{12}$$

$\boxed{\text{z580}}$ $$\mathbb{G}_{3,3}^{(c)}(T_2) = \frac{\sqrt{6}\left(-\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2) + \mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{6}$$

$\boxed{\text{z581}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{21}$$

$\boxed{\text{z582}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{21}$$

$\boxed{\text{z583}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_2,a) = \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1) + \sqrt{5}\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{21}$$

$\boxed{\text{z584}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z585}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z586}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_2,b) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z587}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_2) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,1}^{(b)}(T_2)\right)}{12}$$

$\boxed{\text{z588}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_2) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E)\mathbb{Q}_{2,2}^{(b)}(T_2) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,2}^{(b)}(E)\right)}{12}$$

$\boxed{\text{z589}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_2) = \frac{\sqrt{6}\left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E)\mathbb{Q}_{2,3}^{(b)}(T_2) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_2)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{6}$$

$$\boxed{\text{z590}}\quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_2)=\frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)-9\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)-\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)+9\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)+12\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{84}$$

$$\boxed{\text{z591}}\quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_2)=\frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)+9\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)-\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)-9\mathbb{M}_{3,3}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)+12\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{84}$$

$$\boxed{\text{z592}}\quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_2)=\frac{\sqrt{7}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)-9\mathbb{M}_{3,1}^{(1,-1;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)+9\mathbb{M}_{3,2}^{(1,-1;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)+12\mathbb{M}_{3}^{(1,-1;a)}(A_2)\mathbb{T}_{1,3}^{(b)}(T_1)\right)}{84}$$

$$\boxed{\text{z593}}\quad \mathbb{G}_{2,1}^{(1,0;c)}(T_2,a)=\frac{\sqrt{2}\left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1)+\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(T_1)-\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)+\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z594}}\quad \mathbb{G}_{2,2}^{(1,0;c)}(T_2,a)=\frac{\sqrt{2}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1)+\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,3}^{(b)}(T_1)+\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(T_1)-\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z595}}\quad \mathbb{G}_{2,3}^{(1,0;c)}(T_2,a)=\frac{\sqrt{2}\left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{T}_{1,2}^{(b)}(T_1)-2\mathbb{T}_{2,2}^{(1,0;a)}(E)\mathbb{T}_{1,3}^{(b)}(T_1)+\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$$\boxed{\text{z596}}\quad \mathbb{G}_{2,1}^{(1,0;c)}(T_2,b)=\frac{\sqrt{6}\left(\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)+\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z597}}\quad \mathbb{G}_{2,2}^{(1,0;c)}(T_2,b)=\frac{\sqrt{6}\left(\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{M}_{2,3}^{(b)}(T_2)+\mathbb{T}_{2,3}^{(1,0;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z598}}\quad \mathbb{G}_{2,3}^{(1,0;c)}(T_2,b)=\frac{\sqrt{6}\left(\mathbb{T}_{2,1}^{(1,0;a)}(T_2)\mathbb{M}_{2,2}^{(b)}(T_2)+\mathbb{T}_{2,2}^{(1,0;a)}(T_2)\mathbb{M}_{2,1}^{(b)}(T_2)\right)}{6}$$

$$\boxed{\text{z599}}\quad \mathbb{G}_{3,1}^{(1,0;c)}(T_2)=\frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)-\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)+2\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)-2\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)}{6}$$

$$\boxed{\text{z600}}\quad \mathbb{G}_{3,2}^{(1,0;c)}(T_2)=\frac{-2\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,3}^{(b)}(T_2)+\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(E)-\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)+2\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)}{6}$$

$$\boxed{\text{z601}}\quad \mathbb{G}_{3,3}^{(1,0;c)}(T_2)=\frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(T_2)-\mathbb{G}_{1,2}^{(1,0;a)}(T_1)\mathbb{Q}_{2,1}^{(b)}(T_2)+\mathbb{G}_{1,3}^{(1,0;a)}(T_1)\mathbb{Q}_{2,2}^{(b)}(E)}{3}$$

$$\boxed{\text{z602}}\quad \mathbb{G}_{2,1}^{(1,1;c)}(T_2)=\frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1)+\mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z603}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_1) + \mathbb{M}_{1,3}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

$\boxed{\text{z604}}$ $$\mathbb{G}_{2,3}^{(1,1;c)}(T_2) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_1) + \mathbb{M}_{1,2}^{(1,1;a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_1)\right)}{6}$$

Atomic SAMB

- bra: $\langle s, \uparrow| \,, \langle s, \downarrow|$
- ket: $|s, \uparrow\rangle \,, |s, \downarrow\rangle$

$\boxed{\text{x1}}$ $$\mathbb{Q}_{0}^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$\boxed{\text{x2}}$ $$\mathbb{M}_{1,1}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$\boxed{\text{x3}}$ $$\mathbb{M}_{1,2}^{(1,-1;a)}(T_1) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

$\boxed{\text{x4}}$ $$\mathbb{M}_{1,3}^{(1,-1;a)}(T_1) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

- bra: $\langle s, \uparrow| \,, \langle s, \downarrow|$
- ket: $|p_x, \uparrow\rangle \,, |p_x, \downarrow\rangle \,, |p_y, \uparrow\rangle \,, |p_y, \downarrow\rangle \,, |p_z, \uparrow\rangle \,, |p_z, \downarrow\rangle$

$\boxed{\text{x5}}$ $$\mathbb{Q}_{1,1}^{(a)}(T_1) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x6}}$ $$\mathbb{Q}_{1,2}^{(a)}(T_1) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x7}}$ $\mathbb{Q}_{1,3}^{(a)}(T_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$

$\boxed{\text{x8}}$ $\mathbb{Q}_{1,1}^{(1,0;a)}(T_1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x9}}$ $\mathbb{Q}_{1,2}^{(1,0;a)}(T_1) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x10}}$ $\mathbb{Q}_{1,3}^{(1,0;a)}(T_1) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x11}}$ $\mathbb{G}_{2,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$

$\boxed{\text{x12}}$ $\mathbb{G}_{2,2}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x13}}$ $\mathbb{G}_{2,1}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x14}}$ $\mathbb{G}_{2,2}^{(1,-1;a)}(T_2) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x15}}$ $\mathbb{G}_{2,3}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x16}}$ $\mathbb{G}_{0}^{(1,1;a)}(A_1) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$

$\boxed{\text{x17}}$ $\mathbb{M}_{2,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{6} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$

$\boxed{\text{x18}}$ $\mathbb{M}^{(1,-1;a)}_{2,2}(E) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x19}}$ $\mathbb{M}^{(1,-1;a)}_{2,1}(T_2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x20}}$ $\mathbb{M}^{(1,-1;a)}_{2,2}(T_2) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x21}}$ $\mathbb{M}^{(1,-1;a)}_{2,3}(T_2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x22}}$ $\mathbb{M}^{(1,1;a)}_{0}(A_1) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$

$\boxed{\text{x23}}$ $\mathbb{T}^{(a)}_{1,1}(T_1) = \begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x24}}$ $\mathbb{T}^{(a)}_{1,2}(T_1) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x25}}$ $\mathbb{T}^{(a)}_{1,3}(T_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$

$\boxed{\text{x26}}$ $\mathbb{T}^{(1,0;a)}_{1,1}(T_1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x27}}$ $\mathbb{T}^{(1,0;a)}_{1,2}(T_1) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x28}}$ $\mathbb{T}^{(1,0;a)}_{1,3}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

- bra: $\langle p_x, \uparrow|$, $\langle p_x, \downarrow|$, $\langle p_y, \uparrow|$, $\langle p_y, \downarrow|$, $\langle p_z, \uparrow|$, $\langle p_z, \downarrow|$
- ket: $|p_x, \uparrow\rangle$, $|p_x, \downarrow\rangle$, $|p_y, \uparrow\rangle$, $|p_y, \downarrow\rangle$, $|p_z, \uparrow\rangle$, $|p_z, \downarrow\rangle$

$$\boxed{\text{x29}} \quad \mathbb{Q}_0^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x30}} \quad \mathbb{Q}_{2,1}^{(a)}(E) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$$

$$\boxed{\text{x31}} \quad \mathbb{Q}_{2,2}^{(a)}(E) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x32}} \quad \mathbb{Q}_{2,1}^{(a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x33}}$ $\mathbb{Q}_{2,2}^{(a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x34}}$ $\mathbb{Q}_{2,3}^{(a)}(T_2) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x35}}$ $\mathbb{Q}_{2,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x36}}$ $\mathbb{Q}_{2,2}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$$\boxed{\text{x37}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x38}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x39}} \quad \mathbb{Q}_{2,3}^{(1,-1;a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x40}} \quad \mathbb{Q}_{0}^{(1,1;a)}(A_1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x41}}$ $$\mathbb{G}^{(1,0;a)}_{1,1}(T_1)=\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x42}}$ $$\mathbb{G}^{(1,0;a)}_{1,2}(T_1)=\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x43}}$ $$\mathbb{G}^{(1,0;a)}_{1,3}(T_1)=\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x44}}$ $$\mathbb{M}^{(a)}_{1,1}(T_1)=\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x45}}$ $$\mathbb{M}^{(a)}_{1,2}(T_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x46}}$ $$\mathbb{M}^{(a)}_{1,3}(T_1) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x47}}$ $$\mathbb{M}^{(1,-1;a)}_{3}(A_2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x48}}$ $$\mathbb{M}^{(1,-1;a)}_{1,1}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x49}}\quad \mathbb{M}_{1,2}^{(1,-1;a)}(T_1)=\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x50}}\quad \mathbb{M}_{1,3}^{(1,-1;a)}(T_1)=\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x51}}\quad \mathbb{M}_{3,1}^{(1,-1;a)}(T_1)=\begin{bmatrix} 0 & \frac{\sqrt{5}}{5} & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{5}}{5} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x52}}\quad \mathbb{M}_{3,2}^{(1,-1;a)}(T_1)=\begin{bmatrix} 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{5} & -\frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x53}} \quad \mathbb{M}^{(1,-1;a)}_{3,3}(T_1) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$$

$$\boxed{\text{x54}} \quad \mathbb{M}^{(1,-1;a)}_{3,1}(T_2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x55}} \quad \mathbb{M}^{(1,-1;a)}_{3,2}(T_2) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x56}} \quad \mathbb{M}^{(1,-1;a)}_{3,3}(T_2) = \begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x57}}$ $$\mathbb{M}_{1,1}^{(1,1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$$

$\boxed{\text{x58}}$ $$\mathbb{M}_{1,2}^{(1,1;a)}(T_1) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$$

$\boxed{\text{x59}}$ $$\mathbb{M}_{1,3}^{(1,1;a)}(T_1) = \begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$$

$\boxed{\text{x60}}$ $$\mathbb{T}_{2,1}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x61}}$ $$\mathbb{T}_{2,2}^{(1,0;a)}(E) = \left[\begin{matrix}0 & 0 & - \frac{\sqrt{6}}{6} & 0 & 0 & - \frac{\sqrt{6} i}{12}\\0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6} i}{12} & 0\\- \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12}\\0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0\\0 & - \frac{\sqrt{6} i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0\\\frac{\sqrt{6} i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0\end{matrix}\right]$$

$\boxed{\text{x62}}$ $$\mathbb{T}_{2,1}^{(1,0;a)}(T_2) = \left[\begin{matrix}0 & 0 & 0 & \frac{\sqrt{6} i}{12} & \frac{\sqrt{6}}{12} & 0\\0 & 0 & - \frac{\sqrt{6} i}{12} & 0 & 0 & - \frac{\sqrt{6}}{12}\\0 & \frac{\sqrt{6} i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0\\- \frac{\sqrt{6} i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0\\\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & - \frac{\sqrt{6}}{6}\\0 & - \frac{\sqrt{6}}{12} & 0 & 0 & - \frac{\sqrt{6}}{6} & 0\end{matrix}\right]$$

$\boxed{\text{x63}}$ $$\mathbb{T}_{2,2}^{(1,0;a)}(T_2) = \left[\begin{matrix}0 & \frac{\sqrt{6} i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0\\- \frac{\sqrt{6} i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0\\0 & \frac{\sqrt{6}}{12} & 0 & 0 & - \frac{\sqrt{6}}{12} & 0\\\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12}\\0 & 0 & - \frac{\sqrt{6}}{12} & 0 & 0 & - \frac{\sqrt{6} i}{6}\\0 & 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{6} i}{6} & 0\end{matrix}\right]$$

$\boxed{\text{x64}}$ $$\mathbb{T}_{2,3}^{(1,0;a)}(T_2) = \left[\begin{matrix}\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & - \frac{\sqrt{6}}{12}\\0 & - \frac{\sqrt{6}}{6} & 0 & 0 & - \frac{\sqrt{6}}{12} & 0\\0 & 0 & - \frac{\sqrt{6}}{6} & 0 & 0 & - \frac{\sqrt{6} i}{12}\\0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6} i}{12} & 0\\0 & - \frac{\sqrt{6}}{12} & 0 & - \frac{\sqrt{6} i}{12} & 0 & 0\\- \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6} i}{12} & 0 & 0 & 0\end{matrix}\right]$$

Cluster SAMB

- Site cluster

** Wyckoff: `1a`

$\boxed{\text{y1}}$ $\mathbb{Q}_{0}^{(s)}(A_1) = [1]$

- Bond cluster

** Wyckoff: `3a@3d`

$\boxed{\text{y2}}$ $\mathbb{Q}_{0}^{(b)}(A_1) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

$\boxed{\text{y3}}$ $\mathbb{Q}_{2,1}^{(b)}(E) = \left[-\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}\right]$

$\boxed{\text{y4}}$ $\mathbb{Q}_{2,2}^{(b)}(E) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0\right]$

$\boxed{\text{y5}}$ $\mathbb{T}_{1,1}^{(b)}(T_1) = [i, 0, 0]$

$\boxed{\text{y6}}$ $\mathbb{T}_{1,2}^{(b)}(T_1) = [0, i, 0]$

$\boxed{\text{y7}}$ $\mathbb{T}_{1,3}^{(b)}(T_1) = [0, 0, i]$

** Wyckoff: `6b@3c`

$\boxed{\text{y8}}$ $\mathbb{Q}_{0}^{(b)}(A_1) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y9}}$ $\mathbb{Q}_{2,1}^{(b)}(E) = \left[-\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

$\boxed{\text{y10}}$ $\mathbb{Q}_{2,2}^{(b)}(E) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, 0, 0\right]$

$\boxed{\text{y11}}$ $\mathbb{T}_{1,1}^{(b)}(T_1) = \left[0, 0, \frac{i}{2}, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}\right]$

$\boxed{\text{y12}}$ $\mathbb{T}^{(b)}_{1,2}(T_1) = \left[\frac{i}{2},\, -\frac{i}{2},\, 0,\, 0,\, \frac{i}{2},\, \frac{i}{2}\right]$

$\boxed{\text{y13}}$ $\mathbb{T}^{(b)}_{1,3}(T_1) = \left[\frac{i}{2},\, \frac{i}{2},\, \frac{i}{2},\, -\frac{i}{2},\, 0,\, 0\right]$

$\boxed{\text{y14}}$ $\mathbb{Q}^{(b)}_{2,1}(T_2) = \left[\frac{\sqrt{2}}{2},\, -\frac{\sqrt{2}}{2},\, 0,\, 0,\, 0,\, 0\right]$

$\boxed{\text{y15}}$ $\mathbb{Q}^{(b)}_{2,2}(T_2) = \left[0,\, 0,\, \frac{\sqrt{2}}{2},\, -\frac{\sqrt{2}}{2},\, 0,\, 0\right]$

$\boxed{\text{y16}}$ $\mathbb{Q}^{(b)}_{2,3}(T_2) = \left[0,\, 0,\, 0,\, 0,\, \frac{\sqrt{2}}{2},\, -\frac{\sqrt{2}}{2}\right]$

$\boxed{\text{y17}}$ $\mathbb{M}^{(b)}_{2,1}(T_2) = \left[0,\, 0,\, -\frac{i}{2},\, -\frac{i}{2},\, \frac{i}{2},\, -\frac{i}{2}\right]$

$\boxed{\text{y18}}$ $\mathbb{M}^{(b)}_{2,2}(T_2) = \left[\frac{i}{2},\, -\frac{i}{2},\, 0,\, 0,\, -\frac{i}{2},\, -\frac{i}{2}\right]$

$\boxed{\text{y19}}$ $\mathbb{M}^{(b)}_{2,3}(T_2) = \left[-\frac{i}{2},\, -\frac{i}{2},\, \frac{i}{2},\, -\frac{i}{2},\, 0,\, 0\right]$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	`A`	$\lvert s,\uparrow\rangle$, $\lvert s,\downarrow\rangle$, $\lvert p_x,\uparrow\rangle$, $\lvert p_x,\downarrow\rangle$, $\lvert p_y,\uparrow\rangle$, $\lvert p_y,\downarrow\rangle$, $\lvert p_z,\uparrow\rangle$, $\lvert p_z,\downarrow\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	A	A	[1,2]	[s,p]	[s,p]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'A' (#1) site cluster (1a), 432

SL	position ($\boldsymbol{s}$)	mapping
1	[0.00000, 0.00000, 0.00000]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'A'-'A' [1] (#1) bond cluster (3a@3d), ND, $|\boldsymbol{v}|$= 1.0 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-1.00000, 0.00000, 0.00000]	[0.50000, 0.00000, 0.00000]	[1,-2,-3,4,17,-18,-19,20]	(1,1)	(1,1)	[1,0,0]
2	[0.00000,-1.00000, 0.00000]	[0.00000, 0.50000, 0.00000]	[5,-6,-7,8,13,-14,-15,16]	(1,1)	(1,1)	[0,1,0]
3	[0.00000, 0.00000,-1.00000]	[0.00000, 0.00000, 0.50000]	[9,-10,-11,12,-21,22,23,-24]	(1,1)	(1,1)	[0,0,1]

Table 9: 2-th 'A'-'A' [1] (#2) bond cluster (6b@3c), ND, $|\boldsymbol{v}|$= 1.41421 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000,-1.00000,-1.00000]	[0.00000, 0.50000, 0.50000]	[1,-4,18,-19]	(1,1)	(1,1)	[0,1,1]
2	[0.00000, 1.00000,-1.00000]	[0.00000, 0.50000, 0.50000]	[2,-3,-17,20]	(1,1)	(1,1)	[0,-1,1]
3	[-1.00000, 0.00000,-1.00000]	[0.50000, 0.00000, 0.50000]	[5,-8,-14,15]	(1,1)	(1,1)	[1,0,1]
4	[-1.00000, 0.00000, 1.00000]	[0.50000, 0.00000, 0.50000]	[6,-7,13,-16]	(1,1)	(1,1)	[1,0,-1]
5	[-1.00000,-1.00000, 0.00000]	[0.50000, 0.50000, 0.00000]	[9,-12,21,-24]	(1,1)	(1,1)	[1,1,0]
6	[1.00000,-1.00000, 0.00000]	[0.50000, 0.50000, 0.00000]	[10,-11,-22,23]	(1,1)	(1,1)	[-1,1,0]